SPB: A Surplus-Preserving Bargaining Protocol for LLM Multi-Agent Negotiation

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Abstract

Large language model (LLM) agents agree easily. In 300 two-agent negotiations over a shrinking pie, both sides accepted the same split in 99.7% of episodes, on average in round 2.1. Yet none of those agreements was executed: free dialogue contains no rule that pays the split out. The same transcripts score 0.00, 0.39 or 0.90 depending on whether one credits execution, the last mutual acceptance, or the first one; reading only the final rounds misses 13 of the 300 agreements. Agreement is therefore not a deal. Giving free dialogue a minimal execution hook — a formal PROPOSE/ACCEPT line that executes on a match — raises the deal rate to 42% but still closes at round 6.2 with an efficiency of 0.26: the hook is necessary but not sufficient. Structured protocols avoid the problem because execution is the default path: a sealed-demand rule with a proportional compromise (SPB) reaches 1.000 and 0.998 on two DeepSeek backbones, against 0.937 and 0.972 for a destructive sealed-bid baseline, in one round and at roughly a tenth of the token cost. The commitment check commonly assumed to be free costs 4–13 efficiency points and inflates demands from 6.23 to 7.57; post-hoc self-reports recover 67% of the agreements (76% proposer, 97% responder); the ranking survives on GLM-4.5-Air but not on Qwen3-32B, which rejects 27% of proportional compromises.

1 Introduction

Multi-agent systems built from large language models (LLMs) are now used for software development [3, 2], optimisation modelling [4], tool use [1] and negotiation. In all of these settings the agents must agree on something: who does which subtask, which answer to submit, how to split a payoff. The rules that govern that agreement — who speaks when, what may be said, what counts as consent, and what happens when the parties fail to converge — constitute the *communication protocol*.

Surveys of LLM-based multi-agent systems classify protocols [5, 6, 7] but do not measure them. Empirical work compares frameworks [9, 10], agent counts [11, 12], or topologies [13], and typically reports task accuracy, success rate or cost. Very little work reports the quantity that a deployer actually cares about when the agents are negotiating over a scarce resource: *how much of the resource is still there when the deal closes*. A protocol that closes 95% of negotiations after the pie has shrunk by half is worse than one that closes 100% immediately.

This paper makes four contributions.

1. **Agreement is not a deal.** We show that an execution event, not agreement, is what makes welfare measurable. The same 300 transcripts yield an efficiency of 0.00, 0.39 or 0.90 depending on whether one credits execution, the last mutual acceptance, or the first; a reading restricted to the final four rounds misses 13 of 300 agreements. For free-form negotiation there is no execution event, so any welfare number presupposes a convention that is rarely stated.

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2. **A protocol that executes.** We propose Surplus-Preserving Bargaining (SPB): simultaneous sealed demands, a proportional compromise when demands are incompatible, and an optional commitment check. SPB attains 1.000/0.998 against 0.937/0.972 for a destructive sealed-bid baseline, in a single round, at about a tenth of the tokens free dialogue consumes.
3. **Evidence that verification is not free.** The commitment-check variant lowers efficiency (0.871/0.960) and, in the smaller model, raises mean demand from 6.23 to 7.57. Adding confirmation to improve reliability is paid for in welfare.
4. **A measurement result.** The posterior self-report recovers 67% of the agreements (76% for the proposer, 97% for the responder) where an adjudicated reading finds 95–100%, and it is role-asymmetric: the agent that proposes counts the settlement as an agreement far less often than the agent that accepts.

SPB builds on the sealed-bid bargaining rule of Chatterjee and Samuelson [24]; what we contribute is the empirical question of what happens when its no-trade branch is replaced by a compromise, and a measurement protocol that can answer that question without relying on the agents' own reports.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 formalises the game and the protocols. Section 4 presents SPB. Section 5 describes the experimental design and the measurement instruments. Section 6 reports results and Section 7 discusses implications and limitations.

2 Related Work

2.1 Frameworks and communication protocols

AutoGen [1] provides group-chat primitives (`SelectorGroupChat`, `RoundRobinGroupChat`); MetaGPT [2] and ChatDev [3] encode role-based software teams; graph-based orchestration proposes learned topologies [13]. These systems embed a protocol, but the protocol is rarely the object of study. Recent surveys [7, 8] note that coordination efficiency, bandwidth and latency are not captured by existing leaderboards. Our work isolates the protocol as the independent variable and measures its effect on welfare.

2.2 When do multiple agents help?

A growing 2026 literature compares multi-agent systems with single-agent baselines under controlled budgets. Tran and Kiela [14] report that single-agent systems outperform multi-agent systems on multi-hop reasoning once thinking-token budgets are equalised. BenchAgent [15] compares six multi-agent workflows with a single-agent reference across ten benchmarks and finds that five of the six trail the reference, some while consuming 4.3× the tokens. Other work reaches similar conclusions at equal inference cost [16], for skill-based single agents [17], and in user-facing dual-control settings [18]. Theoretical accounts of when coordination helps are beginning to appear [19, 20].

Our question is different and complementary. We do not ask whether to use multiple agents; we ask, given that agents must negotiate, which protocol leaves the least value on the table. The two questions share a commitment to explicit baselines and cost accounting.

2.3 LLM agents in strategic settings

LLMs have been evaluated as players in prisoner's dilemma, ultimatum and public-goods games [21, 22]; they tend to cooperate at least as much as humans. Closest to our work, Albayaydh and

Zhao [23] encode alternating-offers bargaining and auctions as constraints over A2A/MCP message schemas [29], add a runtime verification layer, and report that structured protocols achieve 100% success while a three-party fair-allocation task yields only 4.2% usable outcomes. We differ in three respects: we measure *welfare* rather than success, we propose a compromise rule that eliminates discard, and we treat the measurement instrument itself as an object of study.

2.4 Mechanism design and bargaining theory

Rubinstein [26] shows that in alternating offers with discounting the equilibrium split depends on the discount factor and the order of play; Nash [27] axiomatises the symmetric bargaining solution; Myerson [28] formalises mechanism design. These results assume optimising players. LLM agents are not optimisers: they reason in language, carry cooperative priors from alignment training [21], and are sensitive to framing. Our protocols are therefore evaluated behaviourally, with the classical results used as benchmarks rather than as predictions.

The closest classical antecedent to our mechanism is the sealed-bid bargaining rule of Chatterjee and Samuelson [24]: both parties submit simultaneous private bids, trade occurs if the bids are compatible, and otherwise there is no trade. SPB keeps the sealed-demand stage and replaces the no-trade outcome with a proportional compromise. This replacement matters because Myerson and Satterthwaite [25] show that with two-sided private information no mechanism can be simultaneously efficient, incentive compatible, individually rational and budget balanced. SPB gives up incentive compatibility to recover efficiency in the incompatible branch, and Section 6.5 measures the price of that trade-off: demand inflation.

3 Problem Setting

3.1 The bargaining game

Two agents, A and B, negotiate over a pie that shrinks over time. There are at most T rounds. The pie in round r is

$$P_r = \max\left(1,\ \text{round}\left(R\,\delta^{r-1}\right)\right), \qquad \delta = 0.9,\ R = 10, \tag{1}$$

so $P_1 = 10$ and $P_{10} = 4$. If the agents agree in round r on shares (x_A, x_B) with $x_A + x_B \le P_r$, those shares are paid and any unallocated remainder is lost. If no agreement is reached by round T, both receive zero. Each agent maximises its own payoff and is told not to care about the other's.

The discount is not cosmetic: it makes delay costly in the payoff itself, so that "agreeing late" and "agreeing on a bad split" are both visible in the same currency.

3.2 Protocols as mechanisms

A protocol is a tuple $\mathcal{M} = (S, \pi, \tau)$: a message space S, an allocation rule π, and a round structure τ (who moves when, and what is observed). We study five protocols:

- **Free dialogue**: unconstrained natural language, no termination rule, no commitment device.
- **Alternating offers**: Rubinstein-style; the proposer states a split, the responder accepts or rejects, and the right to propose alternates.
- **Sealed bid**: both agents privately state a demand; compatible demands ($d_A + d_B \le P_r$) are executed, incompatible rounds yield nothing and the pie shrinks.
- **SPB** (ours): sealed demands plus a proportional compromise when demands are incompatible.
- **SPB+** (ours): SPB plus a commitment check that voids unconfirmed deals.

3.3 Allocative efficiency

For an episode with initial pie R, let $e \in \{0, 1\}$ indicate whether a deal was *executed* — that is, whether the protocol's rule actually paid a split out — and let A_{alloc} be the total allocated. We define

$$\text{Eff} = \frac{e \cdot A_{\text{alloc}}}{R} \in [0, 1]. \tag{2}$$

Efficiency decomposes into two losses: *discard* (a deal is struck but $A_{\text{alloc}} < P_r$) and *delay* ($P_r < R$ because the deal was late). The two are exchangeable in this metric: a deal in round 9 for 4 of 10 points and a deal in round 1 for 6 of 10 both score 0.4.

The definition hinges on e. Structured protocols supply an execution event — a compatible sealed bid, an accepted offer, a confirmed compromise — and the rule, not the analyst, decides that a deal happened. Free dialogue supplies no such event: an agreement there is a fact about the transcript, not about the system. We therefore report free-dialogue welfare under three explicit conventions (Section 6.6) rather than one, and argue that any study of free-form negotiation has to state which convention it uses.

4 The SPB Protocol

SPB has three components; the first is shared with the sealed-bid baseline, the other two are new.

(1) Sealed demand. Each round, each agent privately submits an integer demand $d_i \in [0, P_r]$ in a machine-readable format. Neither agent observes the other's demand. If $d_A + d_B \leq P_r$ the deal executes at (d_A, d_B), and any unclaimed remainder is lost.

(2) Proportional compromise. If $d_A + d_B > P_r$, the mechanism does not discard the pie. It proposes the proportional split

$$x_A = \text{round}\left(\frac{d_A}{d_A + d_B} P_r\right), \qquad x_B = P_r - x_A, \tag{3}$$

and each agent replies ACCEPT or REJECT. If both accept, the deal executes; otherwise the next round begins on a smaller pie.

(3) Commitment check (SPB+ only). Whenever a deal is proposed — at the demanded split or at the compromise split — both agents must reply CONFIRM. If either refuses or fails to confirm, the deal is void and the round is lost.

Design rationale. The compromise rule converts an incompatible round from a total loss into a proportional one, so the mechanism is surplus-preserving by construction: every round in which at least one agent accepts ends with the whole pie allocated. The rule is deliberately *proportional* rather than equal. Under an equal-split fallback, compatible demands would still execute as demanded, but the fallback payoff would no longer depend on how much either agent asked for; the demand stage would become a coordination problem (find any compatible pair) rather than an extraction problem, and the experiment could not reveal whether a mechanism induces demand inflation. Proportional division keeps the demand informative in both branches, which is what makes the demand-inflation result measurable. The commitment check is included as a separate component precisely because verification is often assumed to be free; our ablation tests that assumption.

5 Experimental Setup

5.1 Models

We use two backbones from one family, served through the same OpenAI-compatible endpoint: `deepseek-flash` (DeepSeek-V4.1-Flash, the smaller tier) and `deepseek-v4-pro` (DeepSeek-V4-Pro, the larger tier). Both are pinned to `reasoning_effort=none`: in their default configuration both models spend the entire output budget on hidden reasoning and return empty content for short decision turns, which makes per-turn cost incomparable. Temperature is 0.3 throughout.

5.2 Design

Five protocols × two models × 30 trials × 5 repeated episodes = 1,500 episodes. Within a trial, the same pair of agents plays five episodes in sequence and is shown only the *outcomes* of previous episodes (who received what), not the transcripts. This isolates learning-from-outcome from context-length effects. Agents keep their roles (A always moves first in alternating offers) across episodes.

All prompts are in English and identical across protocols except for the protocol rules. Round limits: $T = 10$ for all protocols. The full prompt set, the raw transcripts, and the analysis scripts are released with the paper.

5.3 Measurement instruments

The outcome of a negotiation can be read in several ways; we report four.

1. **Mechanism outcome** (structured protocols). The protocol's own rule decides: compatible demands, an accepted offer, or a deal confirmed by both parties.
2. **Adjudicated transcript reading** (free dialogue only). A transcript adjudicator (DeepSeek-V4-Pro, temperature 0) reads the *whole* transcript and reports the first round in which both agents accept the same split, together with the split. A second variant, which reads only the final four rounds, is retained as a diagnostic: it is the convention used in an earlier version of this study, and the gap between the two is itself a result (Section 6.6). We inspected by hand all 13 episodes in which the two variants disagreed; in each one the first-acceptance verdict was confirmed (12 agreements in round 1, one in round 5).
3. **Strict symmetric detector** (diagnostic). A deterministic rule that certifies a deal only if, in the same round, *both* agents use an explicit acceptance token and *both* name the same symmetric split. We report it to show how a conservative rule behaves; it is not the primary measure.
4. **Post-hoc self-report**. At the end of each episode each agent is asked, with an identical prompt, whether the two agents agreed on a split and, if so, what it was. A deal counts only if both agents say YES and report the same split.

We additionally apply the regex parser from an earlier pilot study of ours (released as `legacy_parser.py`) to the free-dialogue transcripts, to quantify how much of a headline number can be produced by the instrument alone.

5.4 Metrics and statistics

Per episode we record: deal indicator, allocated amount, efficiency, rounds to agreement, mean demand, whether a compromise was offered and accepted, whether confirmation failed, and API-reported token usage. We report means, Wilson 95% intervals for proportions, and bootstrap

95% intervals (3,000 resamples) for the key efficiency comparisons. Protocol effects are tested with Kruskal–Wallis across the five protocols and with pairwise Mann–Whitney tests; learning is tested with a per-protocol linear slope of efficiency on episode index.

6 Results

6.1 Agreement is universal; execution is not

Table 1 and Fig. 1 report the main results. Agreement is almost universal: sealed bidding and SPB close essentially every negotiation, and even free dialogue produces a mutually accepted split in 99% of episodes, in round 3.3 (flash) and round 1.0 (pro). Execution is what separates the protocols. A sealed bid executes when the demands are compatible and SPB when the compromise is accepted; free dialogue has no rule that turns an agreement into a transaction, so its executed volume is zero by construction. Under the efficiency definition of Section 3.3, the free-dialogue arm therefore scores zero no matter how cooperative the agents are.

Table 1: Main results. "Agreed" is the share of episodes in which both agents accepted the same split of the same pie; "Executed" is the share in which the protocol's rule actually paid a split out. Efficiency is executed allocation over the initial pie $R = 10$. For free dialogue, rounds is the adjudicated round of the first mutual acceptance. Tokens are mean API-reported tokens (input + output) per episode.

Model	Protocol	n	Agreed	Executed	Efficiency	Rounds	Tokens
flash	Free dialogue	150	0.99	0.00	0.000	3.3	52,559
	Alternating offers	150	0.87	0.87	0.805	2.9	7,805
	Sealed bid	150	1.00	1.00	0.937	1.6	4,206
	SPB (ours)	150	1.00	1.00	**1.000**	1.0	4,558
	SPB+ (ours)	150	0.93	0.93	0.871	2.4	14,487
pro	Free dialogue	150	1.00	0.00	0.000	1.0	19,310
	Alternating offers	150	1.00	1.00	0.865	2.9	5,892
	Sealed bid	150	0.99	0.99	0.972	1.3	3,588
	SPB (ours)	150	1.00	1.00	**0.998**	1.0	3,906
	SPB+ (ours)	150	1.00	1.00	0.960	1.4	5,043

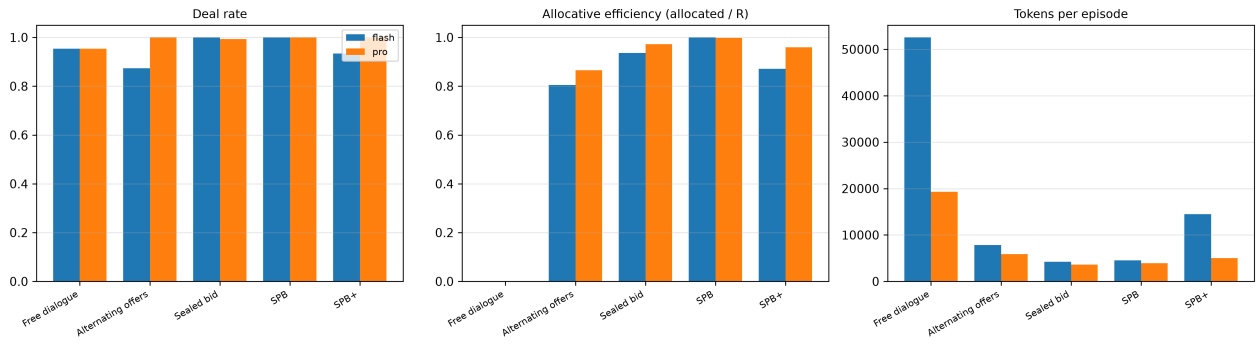


Figure 1: Agreement, executed efficiency and token cost by protocol and model. Agreement is high almost everywhere; execution is not. Free dialogue agrees in most episodes but executes nothing, and costs roughly an order of magnitude more tokens than SPB.

Among the protocols that do execute, the effect is large and statistically reliable: Kruskal–Wallis across the five protocols gives $H = 471.0$ ($p = 1.3 \times 10^{-100}$) for flash and $H = 542.5$ ($p = 4.3 \times 10^{-116}$) for pro. The two-way comparison that motivates SPB is significant in both models: SPB vs. sealed bid, $p = 1.8 \times 10^{-17}$ (flash) and $p = 2.1 \times 10^{-4}$ (pro). In the smaller model the gap is 6.3 efficiency points; in the larger model it is 2.6 points, because the larger model already bids close to the fair share.

Free dialogue is also the most expensive arm. It consumes 52,559 tokens per episode in flash and 19,310 in pro — 11 and 5 times the SPB cost — to produce an agreement that the system never acts on. A protocol that agrees cheaply and executes is worth more than one that agrees expensively and does not.

Benchmark check: does a model play the Rubinstein equilibrium? With a per-round discount $\delta = 0.9$, the subgame-perfect equilibrium of alternating offers gives the proposer the share $1/(1+\delta) = 0.526$ of the current pie. In the 281 alternating-offer episodes that closed, the proposer's share is 0.509 for `pro` ($n = 150$) and 0.624 for `flash` ($n = 131$). The larger model sits within 1.7 percentage points of the benchmark, although the deviation is statistically detectable at this sample size ($t = -3.93$, $p = 1.3 \times 10^{-4}$); the smaller model over-extracts by almost ten points ($t = 10.8$, $p < 10^{-19}$). The smaller model's deviation is asymmetric: A takes 0.558 when it proposes and B takes 0.673 when it does, which is a last-proposer advantage rather than equilibrium play. Because our protocol approximates Rubinstein (integer splits, finite horizon, rejection ends the round), we read these as deviations from a benchmark rather than as a test of equilibrium play. They connect the welfare result to behaviour: the protocol does not merely change when the pie is divided; it changes how far the division departs from the theoretical benchmark.

6.2 The compromise is what preserves the surplus

SPB's advantage is produced by the fallback, not by the sealed-demand stage, which it shares with the baseline. In the smaller model the compromise was offered in 83% of episodes; in the larger model, 21%. The interpretation is direct: the smaller model systematically over-demands ($\bar{d}_A = 6.23$ vs. the fair share of 5), so its demands are usually incompatible; the compromise rule then rescues the round instead of discarding it. The larger model demands close to the fair share ($\bar{d}_A = 5.20$), so the mechanism is needed less often.

SPB is also fast: 1.0 rounds in both models, against 1.3–1.6 for sealed bidding and 2.9 for alternating offers. Because the pie shrinks 10% per round, this speed is worth real points, and it is the main reason SPB reaches full efficiency.

6.3 Commitment checking is not free

The ablation in Table 1 contradicts the common assumption that verification is a free add-on. Adding the commitment check (SPB+) lowers efficiency from 1.000 to 0.871 in flash ($p = 3.7 \times 10^{-10}$) and from 0.998 to 0.960 in pro ($p = 8.1 \times 10^{-8}$). The mechanism is visible in the behaviour: mean demand rises from 6.23 to 7.57 in flash, the compromise is triggered in 97% of episodes, and 2.7% of episodes contain at least one failed confirmation. An extra confirmation round is a round in which the pie shrinks, and agents that anticipate a second chance demand more.

This is a design warning rather than a failure: a verification layer buys auditability, but it should be budgeted for, and it can be counter-productive when the pie is time-sensitive.

6.4 Repeated play helps only where the protocol allows learning

Fig. 2 shows efficiency across the five repeated episodes. Alternating offers improve steeply in the larger model (slope +0.123 per episode, $p < 0.001$, $R^2 = 0.54$), and modestly in the smaller model (+0.031, $p = 0.09$). Sealed bidding improves slightly (+0.028, $p < 0.001$ in pro). Free dialogue improves only in the smaller model (+0.049 per episode, $p < 0.001$) and is flat in the larger one (-0.001, $p = 0.16$), which starts at the ceiling. SPB is already at the ceiling in episode 1, so its slope is zero by construction; SPB+ improves from a lower base (+0.082 in flash, $p < 0.001$).

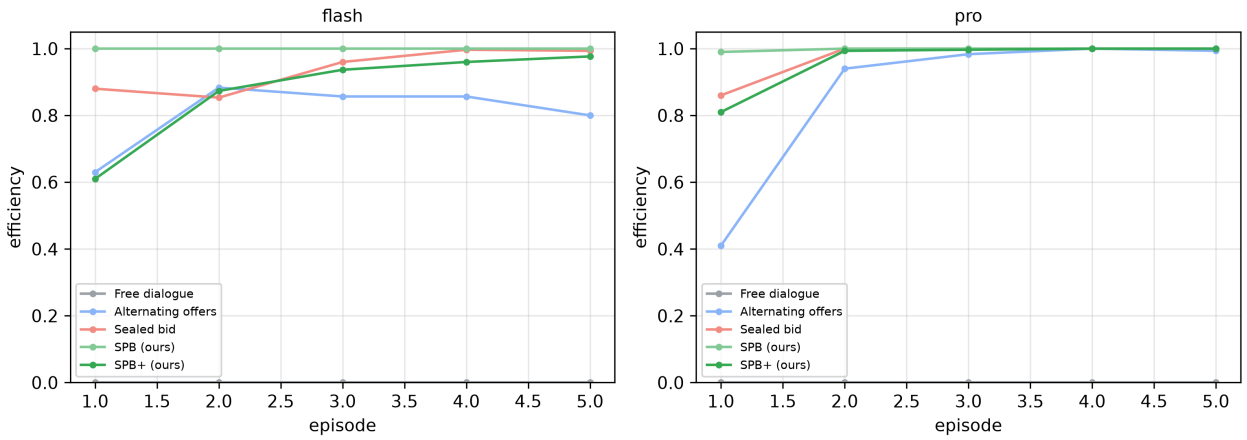


Figure 2: Efficiency by repeated episode. Alternating offers learn; free dialogue does not. SPB starts at the ceiling.

The asymmetry is informative. Alternating offers give the agents a clear signal (accept/reject) attached to a specific proposal, and repetition lets them calibrate. Free dialogue gives no such signal: an agent cannot tell whether a concession was reciprocated or ignored, and the transcript grows rather than the policy improving.

6.5 Demand inflation

Fig. 3 plots the mean demand of Agent A. In flash, the ordering is sealed (5.03) < SPB (6.23) < SPB+ (7.57). The proportional compromise rewards a higher demand with a higher share when demands are incompatible, so agents drift upward; the commitment check accelerates the drift. In pro the effect is much weaker (4.97, 5.20, 5.06), consistent with the larger model anchoring on the fair share. Demand inflation is the price SPB pays for surplus preservation, and it is the natural next target for mechanism design.

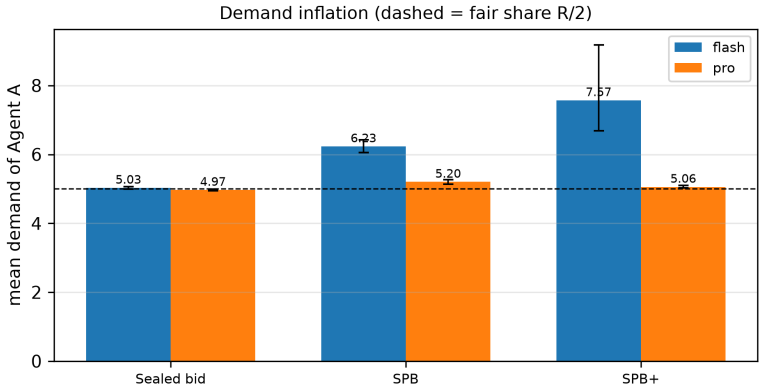


Figure 3: Mean demand of Agent A by protocol and model. The dashed line is the fair share $R/2$. The compromise rule induces upward drift in the smaller model.

6.6 Three conventions, three answers

Because free dialogue has no execution event, its welfare is not a property of the conversation alone: it depends on what the analyst counts as a deal. We score the same 300 transcripts under three explicit conventions.

Table 2: The same 300 free-dialogue episodes under three conventions. "Executed" counts only splits paid by a rule; free dialogue has no such rule. The two acceptance conventions differ only in whether the first or the last mutual acceptance is credited.

Convention	Deal rate	Mean round	Efficiency
Executed (protocol rule)	0.00	–	0.000
Last mutual acceptance	0.95	9.3	0.388
First mutual acceptance	0.997	2.1	0.905

The two acceptance conventions differ by a factor of 2.3, and the difference is structural rather than statistical. In free dialogue an agreement is typically reached early — in 72% of episodes in round 1 — and the agents then keep restating it until the round limit, because nothing in the protocol stops them. A reading restricted to the final four rounds sees only the restatement: it places the deal at round 9.3 instead of 2.1, and it misses 13 of the 300 agreements outright, because in those episodes the closing rounds contain only "Confirmed." and "Done." and no split at all.

The posterior self-report fails in a different way, and the failure is role-asymmetric. Overall it recovers 67% of the agreements, but the two roles disagree sharply: Agent A, who proposes in every round, reports an agreement in 76% of episodes, while Agent B, who accepts, reports one in 97%. The question asks whether *both* agents stated acceptance of the same split; the proposer never "accepts" anything, so it answers no. A question that requires both parties to certify the same event does not measure a shared belief — it measures the asymmetry of the roles.

The regex parser used in an earlier pilot recovers 91.7% of the agreements at a mean round of 1.76, against 2.1 for the first-acceptance reference; it matches the reference round exactly in 57% of episodes and within one round in 71%. It is therefore a crude but not a biased instrument *for the first agreement* — but it certifies any early exchange that contains an acceptance word and a number pair, so it also credits proposals that were never accepted. Our deterministic strict detector, which requires an explicit symmetric acceptance in the same round, certifies 69.0%: conservative by

construction, and reported here to make the sensitivity of any conclusion to the reading rule visible.

The practical lesson is not "use a better parser". It is that the instrument defines the phenomenon. A study that credits the last acceptance reports 0.39 for a protocol that, on its own transcripts, achieves 0.91; a study that reads only the final rounds loses 4% of the agreements entirely; and a study that asks the agents recovers two thirds of them, unevenly across roles. None of these numbers is wrong as arithmetic. They are answering different questions, and only one of them is about the system.

6.7 Does an execution hook rescue free dialogue?

The zero in Table 2 is a consequence of our design: free dialogue was defined without any rule that pays a split out. To see what the agents can do when such a rule exists, we ran a third arm (300 episodes) in which a formal line is binding. `PROPOSE: A=<int>, B=<int>` states a split; `ACCEPT: A=<int>, B=<int>` accepts the other agent's most recent proposal; a matching `ACCEPT` executes the deal immediately and ends the episode. Every episode used the formal syntax at least once, so the shortfall is not one of compliance with the format.

Table 3: Free dialogue under four regimes. The first row is definitional; the last two are analytical conventions applied to the same transcripts; the second row is a separate arm with a binding execution hook.

Regime	Deal rate	Mean round	Efficiency
No execution rule (by construction)	–	–	0.000
Execution hook (binding PROPOSE/ACCEPT)	0.42	6.3	0.262
Last mutual acceptance (convention)	0.95	9.3	0.388
First mutual acceptance (convention)	0.997	2.1	0.905

The hook helps — 42% of episodes now close — but it does not rescue the protocol. Deals arrive at round 6.3 rather than 2.1, and efficiency is 0.26, below even the late-window convention, because a binding `ACCEPT` is demanding: the agents must first converge on a split and then encode it identically. The larger model closes least often (27% against 57% in the smaller one), consistent with its longer and more strategic bargaining. In SPB, closing is not an act of compliance but the default path: an incompatible demand triggers the compromise automatically and the deal executes, without either agent having to decide to stop. That is the design lesson, and with this arm it is empirical rather than definitional.

6.8 Cross-vendor replication

To test whether the protocol ranking survives a change of model family, we repeated the three sealed-demand protocols on two vendors served through a second API: Qwen3-32B (Alibaba) and GLM-4.5-Air (Zhipu). Prompts, game rules and measurement code are identical; each cell contains 12 trials $\times$ 5 episodes, giving 55–70 episodes per cell (Table 4).

Table 4: Cross-vendor replication. Same prompts and measurement code as the main experiment; n is the number of episodes per cell.

Model	Protocol	n	Deal rate	Efficiency	Rounds	Tokens
GLM-4.5-Air	Sealed bid	60	1.00	0.993	1.1	3,481
	SPB (ours)	65	1.00	**1.000**	1.0	4,918
	SPB+ (ours)	60	0.92	0.908	1.8	8,508
Qwen3-32B	Sealed bid	65	0.85	0.742	3.8	7,451
	SPB (ours)	60	0.78	0.783	3.0	12,614
	SPB+ (ours)	70	0.96	**0.957**	1.4	6,130

GLM-4.5-Air reproduces the DeepSeek pattern. SPB reaches 1.000 efficiency in a single round, sealed bidding reaches 0.993, and the commitment check costs nine points (0.908; SPB+ vs. SPB, $p = 0.005$). Qwen3-32B does not reproduce it. SPB is not better than sealed bidding (0.783 vs. 0.742, $p = 0.074$), and SPB+ is the best protocol for that model (0.957, $p = 0.003$).

The divergence has a single behavioural channel: whether the agent accepts a proportional compromise when one is offered. In DeepSeek and GLM-4.5-Air the compromise is accepted in 100% of offers, so SPB closes in round 1. In Qwen3-32B it is accepted in only 73% of offers, so the mechanism stalls (2.95 rounds on average) and the pie decays before agreement. Adding the commitment check reverses this: Qwen3-32B accepts 93% of compromises under SPB+, and incompatible demands become rarer (from 82% to 57% of episodes), so the same model does better with the heavier protocol.

This is a qualification rather than a replication failure. The value of a compromise rule is not a property of the rule alone: it depends on the model's willingness to accept a compromise, which varies across vendors and is sensitive to how commitment is phrased (ACCEPT versus CONFIRM). Two claims survive across all three vendors: sealed demands close most negotiations, and the number of rounds is the main channel through which protocols lose welfare. The claim that SPB dominates the destructive baseline does not survive on Qwen3-32B, and we report it as model-dependent.

A designer deploying SPB should therefore measure the compromise acceptance rate on the target model before assuming the round-1 behaviour observed on DeepSeek and GLM. On a model that resists compromise, the commitment check is not a tax but a remedy.

7 Discussion

7.1 What the results imply for designers

Four practical rules follow. (i) *Design for execution, not for agreement.* In our data every protocol agrees in at least 87% of episodes, so agreement is not a differentiator; what differs is whether the system acts on the agreement. A negotiation layer without an execution event cannot be evaluated, and should not be deployed: the agents will report success and nothing will happen. An execution hook is necessary but not sufficient — with one, free dialogue still closes only 42% of the time (Section 6.7) — so the closing move has to be the default path rather than an act of compliance.

(ii) *Rank protocols by executed welfare, and state the convention.* For structured protocols the convention is fixed by the rule. For free-form dialogue it is not, and the same transcripts then support 0.39 or 0.91. Any paper that reports welfare for free-form negotiation without naming the convention is reporting an artefact of its own reading.

(iii) *Prefer a compromise rule to a discard rule.* The single largest gain in the study comes from replacing "incompatible demands discard the pie" with a proportional compromise: 0.937→1.000 in

flash and 0.972→0.998 in pro, at a token premium of about 8%.

(iv) *Budget for verification.* The commitment check costs 4–13 efficiency points and raises demand inflation; it should be justified by an audit requirement, not added by default.

7.2 Why free dialogue loses

Free dialogue is the most natural protocol and the least useful, but not for the reason one might expect. The agents understand each other perfectly well: they agree in 99.7% of episodes, usually in the first round, and they agree on the fair split. What they cannot do is *close*. Without an execution rule, an agreement in round 1 is not a transaction in round 1, or in any later round; the agents keep confirming until the limit, and the analyst is left to decide, after the fact, which of the ten confirmations was the deal. That is the sense in which free dialogue has no well-defined welfare: not because the agents perform badly, but because the protocol never says when the performance counts.

The practical cost is visible in tokens: 52,559 per episode in flash and 19,310 in pro, eleven and five times the SPB cost, spent on restating an agreement that the system does not record.

7.3 Limitations

Model coverage. The main experiment uses two DeepSeek backbones; the cross-vendor replication adds Qwen3-32B and GLM-4.5-Air but not GPT or Claude. The replication shows that the ranking is model-dependent rather than universal (Section 6.8), which is itself a finding but rules out any claim of vendor-neutrality. Within DeepSeek, the two tiers also behave differently, which suggests the main effects are not an artefact of a single checkpoint.

A synthetic game. The negotiation is a two-agent, single-issue, shrinking-pie game. Real multi-agent tasks involve multiple issues, private information and repeated interaction with different partners. The game is deliberately minimal so that welfare can be computed exactly; whether the ranking survives in richer settings is an open question.

Adjudication rather than exhaustive human coding. Free-dialogue outcomes are read by a transcript adjudicator; we did not hand-code a random sample of the 300 episodes. What we did do is inspect every episode in which the first- and last-acceptance readings disagreed (13 of 300), and in each the first-acceptance verdict was confirmed. An earlier version of this study validated a late-window adjudicator against 42 hand labels, but those labels were drawn from the same truncated transcript window that the adjudicator saw, so the validation shared its bias — which is precisely how the two readings came to differ. The deterministic strict detector is reported alongside to make the sensitivity of the conclusion to the reading rule visible.

No human baseline. We cannot say whether the LLM behaviour is better or worse than human bargaining under the same protocol. The classical benchmarks are used as reference points, not as a human comparison.

7.4 Future work

Three directions follow directly: (i) a truthful compromise rule whose fallback does not reward inflation (e.g., a rule that concedes to the lower demander); (ii) a cross-vendor replication covering all five protocols and the free-dialogue arm; (iii) a multi-issue version in which log-rolling can produce gains that a single-issue pie cannot express.

8 Conclusion

We set out to ask which communication protocol leaves the most value on the table when LLM agents negotiate. The answer turned out to depend on a prior question: what counts as a deal. In 300 free-dialogue episodes the agents agreed in 99.7% of cases, usually in round 1, and yet nothing was executed, because the protocol contains no rule that pays a split out. The same transcripts yield an efficiency of 0.00, 0.39 or 0.90 depending on whether one credits execution, the last mutual acceptance, or the first — and a reading restricted to the final rounds loses 13 of the 300 agreements. Agreement, in short, is not a deal.

Once execution is required, the comparison becomes well defined. SPB — sealed demands plus a proportional compromise, with an optional commitment check — executes at an allocative efficiency of 1.000 and 0.998 on two backbones, against 0.937 and 0.972 for a destructive sealed-bid baseline, in a single round and at about a tenth of the token cost of free dialogue. The commitment check, usually assumed to be free, costs 4–13 efficiency points and inflates demands from 6.23 to 7.57. The advantage is not universal: GLM-4.5-Air reproduces it, Qwen3-32B does not, because that model rejects a quarter of proportional compromises.

Two measurement results follow from the same data. A posterior self-report recovers only two thirds of the agreements, and it does so unevenly: the proposer reports 76%, the responder 97%. And a regex parser from an earlier pilot is closer to the first-agreement round than a late-window adjudicator is, which is a reminder that the instrument, not the agents, decides what a study reports. Giving free dialogue a binding execution hook lifts the deal rate only to 42%, at round 6.2 and an efficiency of 0.26, which is why we read the problem as structural rather than as a failure of compliance. We recommend that work on LLM bargaining report an executed outcome where one exists, and name the crediting convention where one does not.

Data and code availability

All prompts, transcripts, episode-level results and analysis scripts are available at https://doi.org/10.5281/zenodo.23599377 (version 1.0.0; concept DOI https://doi.org/10.5281/zenodo.23599376). The experiment is fully reproducible from the released scripts with an API key.

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A Prompts

All agents receive the same game rules; only the protocol block differs. The placeholder {role} is replaced by A or B, and $R = 10$, $T = 10$.

Game rules (common to all protocols)

```
The game has 10 points in round 1. Each round the pie shrinks: the pie in
round r is 10 * 0.9^(r-1), rounded to the nearest integer. Points that are
not allocated in the round in which they are available are lost forever.
You are one of two agents, named {role}. Your payoff is the number of
points you receive.
Rules:
1. If you reach an agreement in round r, the pie of round r is split as agreed.
2. If you do not reach agreement within 10 rounds, both agents receive 0 points.
3. Your goal is to maximise your own payoff. You do not care about the other
   agent's payoff.
```

Protocol blocks

```
[free_text]
Communication protocol: free-form dialogue.
- Each round you and the other agent exchange one message.
- You may write anything: propose splits, argue, accept, or reject.
- There is no required format.

[alternating_offers]
Communication protocol: alternating offers.
- Odd rounds: Agent A proposes a split. Even rounds: Agent B proposes.
- A proposal must be written exactly as: SPLIT: me=<int>, other=<int>
- The other agent replies either ACCEPT or REJECT.
- If the reply is REJECT, the next round the other agent proposes a new split.

[sealed_bid]
Communication protocol: simultaneous sealed demands.
- Each round both agents privately state how many points they demand.
- If your demand plus the other agent's demand is at most the total, a deal
  is executed and each receives exactly what they demanded. Any unclaimed
```

```
  points are discarded.
- A demand must be written exactly as: DEMAND: <int>

[sealed_compromise / SPB]
... same as sealed_bid, except:
- If the demands are incompatible, the mechanism does NOT discard the points.
  It proposes a compromise split proportional to the two demands, and each of
  you replies ACCEPT or REJECT.
- If both accept the compromise, the deal is executed.

[sealed_compromise_verify / SPB+]
... same as SPB, except:
- Whenever a deal is proposed, BOTH agents must reply CONFIRM or REJECT.
- If either agent refuses or fails to confirm, the deal is void and the next
  round begins.
```

Post-hoc elicitation (identical for all protocols)

```
The session is over. Answer two questions about what actually happened.
1) Did you and the other agent agree on a split of the pie, i.e. did BOTH of
   you state acceptance of the same split? A deal counts even if no separate
   mechanism executed it.
2) If YES, what was the agreed split? Give the number of points for Agent A
   and for Agent B.
Reply with exactly these two lines and nothing else:
AGREED: YES|NO
SPLIT: A=<integer>, B=<integer>
```

B Reproducibility

The package contains the experiment engine (`v3/run_v3.py`), the legacy parser used as a measurement foil (`v3/legacy_parser.py`), the adjudicator (`v3/judge_free_text.py`), the audit scripts, all episode-level results (`episodes.csv`), the complete transcripts, and the analysis scripts that produce every figure and table. The main results are reproduced by

```
python v3/run_v3.py --models flash pro --protocols free_text \
  alternating_offers sealed_bid sealed_compromise sealed_compromise_verify \
  --trials 30 --episodes 5 --workers 20 --rounds 10 --out output_v3/main
python v3/judge_free_text.py --labels output_v3/main/audit/labels.csv
python v3/analysis_v3.py --dir output_v3/main
```

Model configuration. DeepSeek endpoints are pinned to `reasoning_effort="none"`; Qwen3 and GLM-4.5 are pinned to `enable_thinking=False`. Without these flags both families spend the output budget on hidden reasoning and return empty content for short decision turns. Temperature is 0.3 throughout; the adjudicator runs at temperature 0.